

Partial Derivatives and Differentiability Worksheet

1. Consider the surface given by the function $f(x, y) = \sin(xy) + x$.

a) What is the equation of the curve C_1 which is the intersection between $f(x, y)$ and $x = 2$?

The curve C_1 is given by $f(2, y) = \sin(2y) + 2 \rightarrow C_1 = f(y) = \sin(2y) + 2$

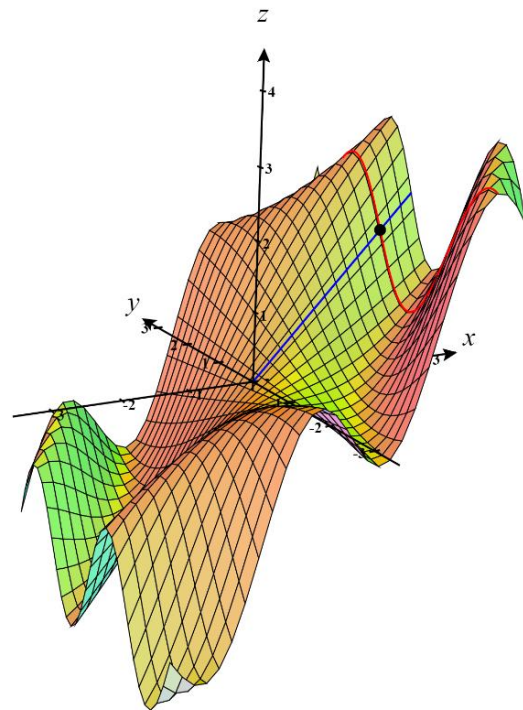
This curve is represented by red curve in the picture below.

b) Find the slope of the tangent line to the curve C_1 at the point $(2, 0, 2)$.

We calculate that $f'(y) = 2\cos(2y)$ and so

$$f'(0) = 2\cos(2(0)) = 2(1) = 2.$$

Therefore the slope of the tangent line to the curve C_1 at the point $(2, 0, 2)$ is $f'(0) = 2$.



c) What is the equation of the curve C_2 which is the intersection between $f(x, y)$ and $y = 0$?

The curve C_2 is given by $f(x, 0) = \sin(x(0)) + x \rightarrow C_2 = f(x) = \sin(0) + x = x$

This curve is represented by the blue curve in the picture below.

d) Find the slope of the tangent line to the curve C_2 at the point $(2, 0, 2)$.

We calculate that $f'(x) = 1$ and so $f'(2) = 1$.

Therefore the slope of the tangent line to the curve C_2 at the point $(2, 0, 2)$ is $f'(2) = 1$.

2. Determine the signs of the partial derivatives for the function f whose graph is shown below.

a) $f_x(1,2)$

We look at the red dashed line in the top picture.

We see that the z -coordinate increases as x increases so $f_x(1,2) > 0$.

b) $f_y(1,2)$

We look at the solid red line in the top picture.

We see that the z -coordinate decreases as y increases so $f_y(1,2) < 0$.

c) $f_x(-1,2)$

We look at the blue dashed line in the top picture.

We see that the z -coordinate decreases as x increases so $f_x(-1,2) < 0$.

d) $f_y(-1,2)$

We look at the solid blue line in the top picture.

We see that the z -coordinate decreases as x increases so $f_y(-1,2) < 0$.

e) $f_{xx}(-1,2)$

We look at the blue dashed curve in the bottom picture.

We see that the curve at this point is concave up and so $f_{xx}(-1,2) > 0$.

f) $f_{yy}(-1,2)$

We look here at the solid blue curve in the bottom picture.

We see that the curve at this point is concave down and so $f_{yy}(-1,2) < 0$.

g) $f_{xy}(1,2)$

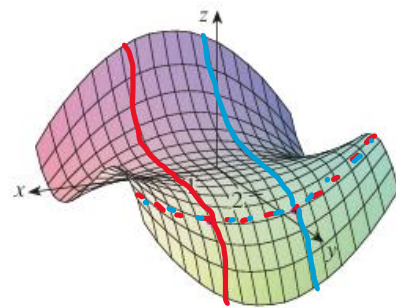
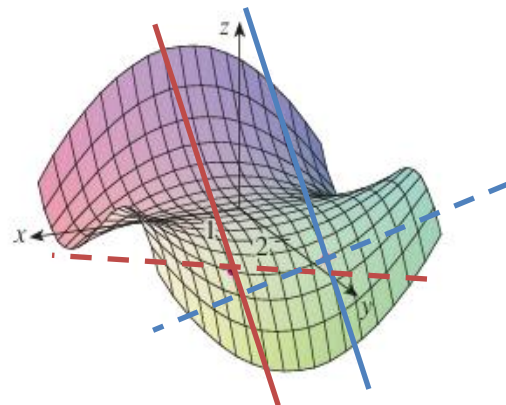
We look at the red dashed curve and consider the tangent line with respect to x at this point. We see that here $f_x(1,2) > 0$. Now, f_{xy} represents how much f_x changes as we move in the positive y direction.

We see that if we move slightly over (say to a point like $(1,2.1)$ moving down the solid red curve) the slope of the tangent line (with respect to x) will be steeper and therefore f_x is increasing with respect to y . Therefore, $f_{xy}(1,2) > 0$.

h) $f_{yx}(-1,2)$

We look at the blue solid curve and consider the tangent line with respect to y at this point. We see that here $f_y(-1,2) < 0$. Now, f_{yx} represents how much f_y changes as we move in the positive x direction.

We see that if we move slightly over (say to a point like $(-0.9,2)$ moving down the dash blue curve) the slope of the tangent line (with respect to y) will be steeper (but in a negative direction) and therefore f_y is decreasing with respect to x . Therefore, $f_{yx}(-1,2) < 0$.



3. Find all the first partial derivatives of the function.

a) $f(x, y) = y^5 - 3xy$

$$f_x = -3y$$

$$f_y = 5y^4 - 3x$$

b) $f(x, y) = \sqrt{x} \ln(y)$

$$f_x = \frac{1}{2} x^{-1/2} \ln(y)$$

$$f_y = \frac{\sqrt{x}}{y}$$

c) $f(x, y) = \tan(xy)$

$$f_x = \sec^2(xy) y$$

$$f_y = \sec^2(xy) x$$

d) $f(x, y) = \frac{x}{(x+y)^2}$

$$f_x = \frac{(x+y)^2(1) - 2(x+y)(1)(x)}{(x+y)^4} = \frac{x+y-2x}{(x+y)^3} = \frac{y-x}{(x+y)^3}$$

$$f_y = \frac{(x+y)^2(0) - 2(x+y)(1)(x)}{(x+y)^4} = \frac{-2x}{(x+y)^3}$$

e) $f(x, y) = x \cos(y-x)$

$$f_x = \cos(y-x) - x \sin(y-x)(-1) = \cos(y-x) + x \sin(y-x)$$

$$f_y = -x \sin(y-x)$$

f) $w = \ln(x + 2y + 3z)$

$$w_x = \frac{1}{x + 2y + 3z} (1) = \frac{1}{x + 2y + 3z}$$

$$w_y = \frac{1}{x + 2y + 3z} (2) = \frac{2}{x + 2y + 3z}$$

$$w_z = \frac{1}{x + 2y + 3z} (3) = \frac{3}{x + 2y + 3z}$$

g) $w = ze^{xyz}$

$$w_x = ze^{xyz} (yz) = yz^2 e^{xyz}$$

$$w_y = ze^{xyz} (xz) = xz^2 e^{xyz}$$

$$w_z = ze^{xyz} (xy) + e^{xyz} = xyze^{xyz} + e^{xyz} = e^{xyz} (xyz + 1)$$

h) $F(x, y) = \int_y^x \cos(e^t) dt$ (Hint: Use the Fundamental Theorem of Calculus).

$$F_x(x, y) = \cos(e^x)$$

To find $F_y(x, y)$ we first rewrite $F(x, y) = \int_y^x \cos(e^t) dt = -\int_x^y \cos(e^t) dt$.

Therefore, $F_y(x, y) = -\cos(e^y)$.

4. Find all the second partial derivatives of $f(x, y) = x^3y^5 + 2x^4y$.

We see that $f_x = 3x^2y^5 + 8x^3y$ & $f_y = 5x^3y^4 + 2x^4$

Therefore,

$$f_{xx} = 6xy^5 + 24x^2y$$

$$f_{xy} = 15x^2y^4 + 8x^3$$

$$f_{yy} = 20x^3y^3$$

$$f_{yx} = 15x^2y^4 + 8x^3$$

5. Find all second partial derivatives of $f(x, y) = \sin^2(2x + 3y)$.

We see that

$$f_x = 2 \sin(2x + 3y) \cos(2x + 3y)(2) = 4 \sin(2x + 3y) \cos(2x + 3y)$$

$$f_y = 2 \sin(2x + 3y) \cos(2x + 3y)(3) = 6 \sin(2x + 3y) \cos(2x + 3y)$$

Therefore,

$$f_{xx} = 4 \sin(2x + 3y) [-\sin(2x + 3y)(2)] + 4 \cos(2x + 3y)(2) \cos(2x + 3y) = -8 \sin^2(2x + 3y) + 8 \cos^2(2x + 3y)$$

$$f_{xy} = 4 \sin(2x + 3y) [-\sin(2x + 3y)(3)] + 4 \cos(2x + 3y)(3) \cos(2x + 3y) = -12 \sin^2(2x + 3y) + 12 \cos^2(2x + 3y)$$

$$f_{yy} = 6 \sin(2x + 3y) [-\sin(2x + 3y)(3)] + 6 \cos(2x + 3y)(3) \cos(2x + 3y) = -18 \sin^2(2x + 3y) + 18 \cos^2(2x + 3y)$$

$$f_{yx} = 6 \sin(2x + 3y) [-\sin(2x + 3y)(2)] + 6 \cos(2x + 3y)(2) \cos(2x + 3y) = -12 \sin^2(2x + 3y) + 12 \cos^2(2x + 3y)$$

6. Find $\frac{\partial z}{\partial x}$ given that $e^{xyz} = 2$.

$$\text{First we note that } e^{xyz} = 2 \Rightarrow xyz = \ln(2) \Rightarrow z = \frac{\ln(2)}{xy}.$$

$$\text{Then we calculate that } \frac{\partial z}{\partial x} = \frac{xy(0) - \ln(2)y}{x^2 y^2} = \frac{-\ln(2)}{x^2 y}$$

7. Find the value of $\frac{\partial z}{\partial x}$ at the point $(-1, 2, 1)$ if the equation $xy - z^3 x + 5yx = 9$ defines z as a function of the two independent variables x and y and the partial derivative exists.

We begin by calculating the partial derivative of both sides with respect to x , while treating y as a constant and z as a mystery function of x and y .

$$\frac{\partial}{\partial x} [xy - z^3 x + 5yx] = \frac{\partial}{\partial x} [9] \Rightarrow y - \left(z^3 + 3xz^2 \frac{\partial z}{\partial x} \right) + 5y = 0$$

$$\Rightarrow 6y - z^3 - 3xz^2 \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow 6y - z^3 = 3xz^2 \frac{\partial z}{\partial x}$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{6y - z^3}{3xz^2}$$

Therefore, $\left. \frac{\partial z}{\partial x} \right|_{(x,y,z)=(-1,2,1)} = \frac{6(2)-(1)^3}{3(-1)(1)^2} = \frac{11}{-3}$

8. Suppose that $z = f(x, y)$ is determined implicitly by the equation $x^2 - y^2 + z^2 + 2x \sin(y - xz) = 3$. Find $\frac{\partial z}{\partial y}$ and its value at the point $(0, 1, 2)$.

We begin by calculating the partial derivative of both sides with respect to y , while treating x as a constant and z as a mystery function of x and y .

$$\begin{aligned} \frac{\partial}{\partial y} [x^2 - y^2 + z^2 + 2x \sin(y - xz)] &= \frac{\partial}{\partial y} [3] \Rightarrow -2y + 2z \frac{\partial z}{\partial y} + 2x \cos(y - xz) \left(1 - x \frac{\partial z}{\partial y} \right) = 0 \\ &\Rightarrow -2y + 2z \frac{\partial z}{\partial y} + 2x \cos(y - xz) - 2x^2 \cos(y - xz) \frac{\partial z}{\partial y} = 0 \\ &\Rightarrow 2x^2 \cos(y - xz) \frac{\partial z}{\partial y} - 2z \frac{\partial z}{\partial y} = -2y + 2x \cos(y - xz) \\ &\Rightarrow \frac{\partial z}{\partial y} = \frac{-2y + 2x \cos(y - xz)}{2x^2 \cos(y - xz) - 2z} \end{aligned}$$

Therefore, know that

$$\left. \frac{\partial z}{\partial y} \right|_{(0,1,2)} = \frac{-2(1) + 2(0) \cos(1 - (0)(2))}{2(0)^2 \cos(1 - (0)(2)) - 2(2)} = \frac{-2 + 0}{0 - 4} = \frac{1}{2}$$

9. Consider the equation $yz + x \ln(y) = z^2$. Find $\frac{\partial z}{\partial x}$.

$$\begin{aligned} \frac{\partial}{\partial x} [yz + x \ln(y)] &= \frac{\partial}{\partial x} [z^2] \\ y \frac{\partial z}{\partial x} + \ln(y) &= 2z \frac{\partial z}{\partial x} \\ y \frac{\partial z}{\partial x} - 2z \frac{\partial z}{\partial x} &= -\ln(y) \\ \frac{\partial z}{\partial x} (y - 2z) &= -\ln(y) \\ \frac{\partial z}{\partial x} &= \frac{-\ln(y)}{(y - 2z)} \end{aligned}$$

10. Suppose that for the function $f(x, y)$ we know that f_{xy} and f_{yx} are both differentiable at $(x, y) = (2, 5)$. Prove that $f_{xy}(2, 5) = f_{yx}(2, 5)$.

Since we know that f_{xy} and f_{yx} are both differentiable at $(x, y) = (2, 5)$ we can say that they are both continuous at $(x, y) = (2, 5)$. However, we know from section 3.3 that if the mixed partials of a function are continuous at a point they must be equal at that point.

Therefore, we can conclude that $f_{xy}(2, 5) = f_{yx}(2, 5)$.